

RAJAT MITTAL

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Education

Indian Institute of Technology (IIT), Jammu

M.Tech. in Computer Science & Engineering

2020 - 2023

8.67/10 CGPA

Work Experience

Texas Instruments

Jul 2025 – Present

Software Development Engineer II (SDE 2)

Bengaluru, IN

- Driving architecture of NextGen application to transform legacy test solutions software for scalability and maintainability.

Software Development Engineer I (SDE 1)

Sept 2024 – Jul 2025

- Optimized Testware architecture to scale execution across semiconductor manufacturing stages.
- Refactored legacy codebase to improve maintainability and reduce operational risks.
- Skills:** C, C++, Tcl/Tk, Python, Oracle, Code Reviews, Data Structures, SQL, Git

Precisely Software & Data Ltd

Aug 2023 – Sep 2024

Software Engineer - ML

Bengaluru, IN

- Developing global automated ML-driven processes for deduplicating over **100Mn** geo-addresses across countries using Python and ML, enhancing data quality and consistency.
- Establishing scalable AWS data pipelines to deploy and monitor the ML pipeline, ensuring robust data quality assurance and operational efficiency.
- Created a natural language query tool with Langchain and LLMs for data retrieval from multiple geolocation APIs.
- Skills:** Python, Data Structures, Code Review, ML, SQL, AWS, Git

Amazon Development Center

Feb 2023 – Jul 2023

Applied Scientist Intern

Bengaluru, IN

- Developed and optimized a ranking model for Amazon's High-Velocity Events, forecasting key business metrics 60-90 days ahead and incorporating guardrails to filter low-quality product ASINs.
- Integrated Amazon's M5 and SBERT embeddings into the model, improving data processing pipelines and achieving a **39%** reduction in Mean Absolute Error (MAE) across various MENA regions.
- Skills:** Python, PyTorch, AWS, SQL, Sagemaker, PySpark, EMR for large-scale data processing.

Zomato Limited

Aug 2022 – Nov 2022

Tech Intern

Gurgaon, IN

- Leveraged reinforcement learning to deliver personalized food/grocery ordering recommendations.
- Designed and managed a robust system architecture to support the delivery of recommendations to a user base of over 50 million, ensuring scalability and reliability.
- Skills:** Python, Data Structures, SQL, Reinforcement Learning, System Design

Ericsson Global (R&D) GAIA

Jul 2021 – Dec 2021

Tech Intern

Bengaluru, IN

- Spearheaded the development of an automation tool to identify faults and errors in cell towers using ML.
- Achieved a **200x** improvement in efficiency by implementing DAGs for fault detection, minimizing human effort.

Projects

A Linux Shell in C language

C, Linux, Signal Handling, Command Parsing, Job Control, Redirection, Piping

- Implemented a comprehensive Linux shell with support for standard and custom commands, including cd, ls, mkdir, and others.
- Developed advanced features such as redirection, piping, and job control, along with robust error and signal handling.
- Maintained command history across sessions, ensuring persistent user experience.

Temporal Action Localization in Untrimmed Videos

Prof. Badri Subudhi, IIT Jammu

Python, PyTorch, Computer Vision, Object Detection, Transformers

- Designed and introduced the Temporal Averager for Action Localization (TAAL) module, enhancing feature extraction from nearby clip embeddings.
- Accomplished better performance over existing state-of-the-art methods on benchmark datasets, demonstrating the effectiveness of the proposed approach.

Technical Skills

Programming Languages: C, C++, Python, JavaScript

Backend: Node.js, Express.js, Django, Flask

Frontend: HTML, CSS, Bootstrap, React JS, Tailwind CSS

Databases & Clouds : SQL, MongoDB, PostgreSQL, Oracle, PySpark, AWS

Developer Tools: Git, Jupyter, Postman, Linux, Windows, VS Code, Docker, Kubernetes, Tcl/Tk

Coursework: Data Structures and Algorithms, Operating Systems, Databases, Machine Learning

Achievements

- Solved over 1500 programming problems on platforms including Leetcode, HackerEarth, Codechef, Codeforces etc.
- Achieved the highest academic performance in M.Tech. Computer Technology at IIT Jammu.
- Scored **100% in PCM** (Physics, Chemistry, Maths) in Senior School board examinations.